



Introduction to Accounting and Finance Analytics

Acctg & Fin | BUS AN 500 | Class 01
Spring Qtr | 2026

FOSTER
SCHOOL OF BUSINESS

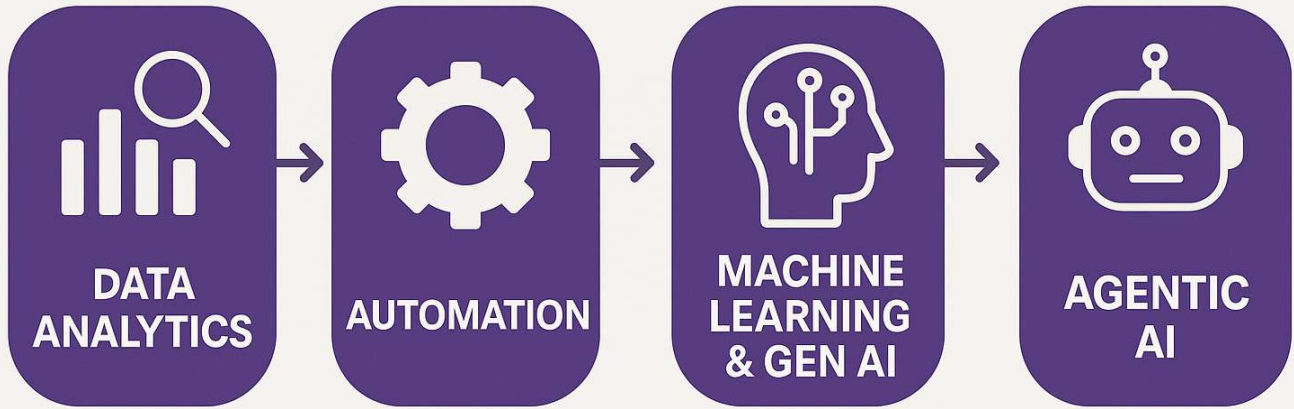
 UNIVERSITY of WASHINGTON

What will we do today?

- **Course Overview**
- Introduction to Accounting & Finance Analytics
 - Define the scope of the Accounting & Finance Function
 - Analyze Transaction Level Data
 - Analyze Public Disclosures
 - Provide a Solution to Disclosure Decision



Course Diagram



Course Overview

Accounting and Finance

Business analytics is transforming how organizations understand and execute accounting and finance processes. In this course, we examine how data, automation, and artificial intelligence come together to move from insight generation to intelligent, scalable workflows.

Business Analytics for Accounting and Finance examines the evolving role of data, automation, and artificial intelligence in modern financial and accounting environments. The course investigates how organizations transition from data-driven insight generation to fully integrated automation systems, where processes are increasingly executed, monitored, and adapted by intelligent technologies rather than human intervention alone.



Syllabus

Read me first! The course [Syllabus](#).



Course Overview

An overview of the classes in BUS AN 500.



Deliverables

Information on deliverables and links to the submission portals.



Key Terms

Definitions for Key Terms used in BUS AN 500.



FAQs

A repository of Frequently Asked Questions.



Video Materials

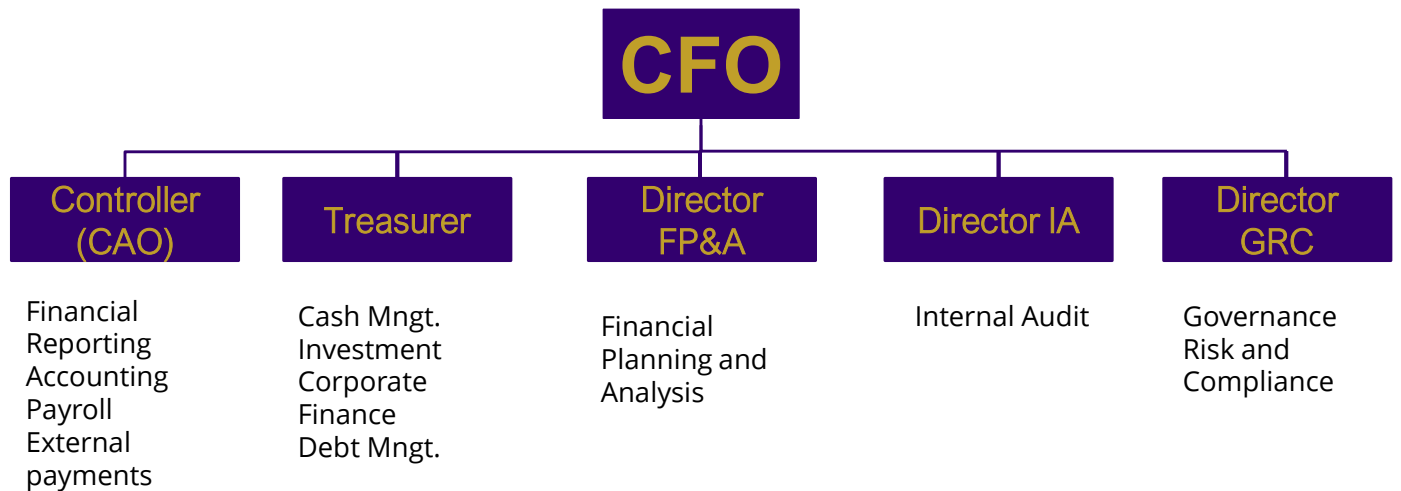
Additional video and other materials.

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Accounting & Finance Function



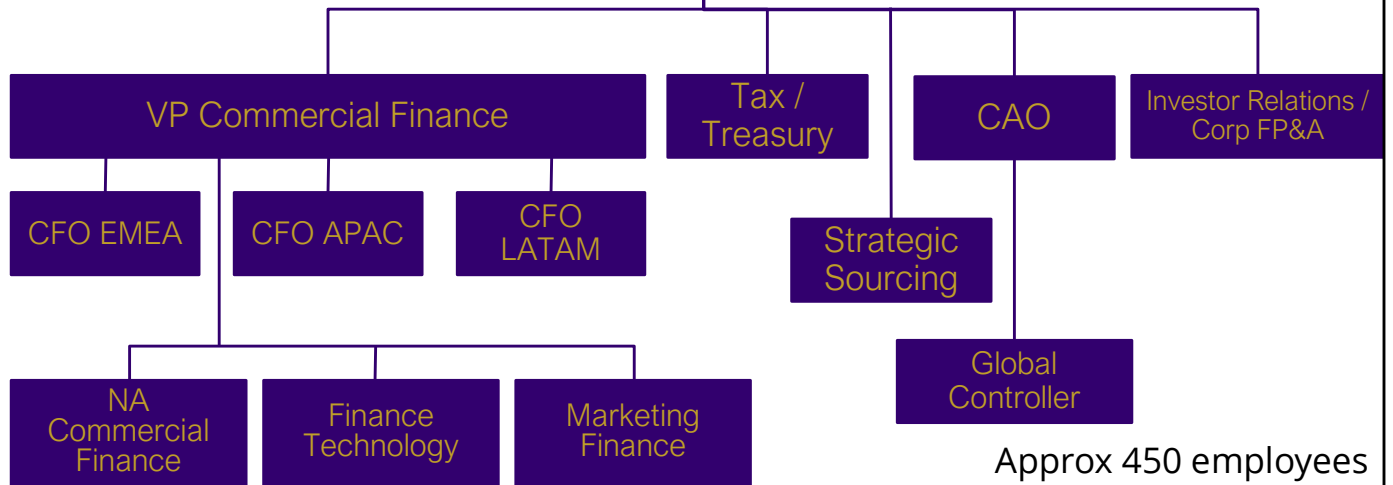
A now Public Company Example (pre-IPO)



Approx 120 employees

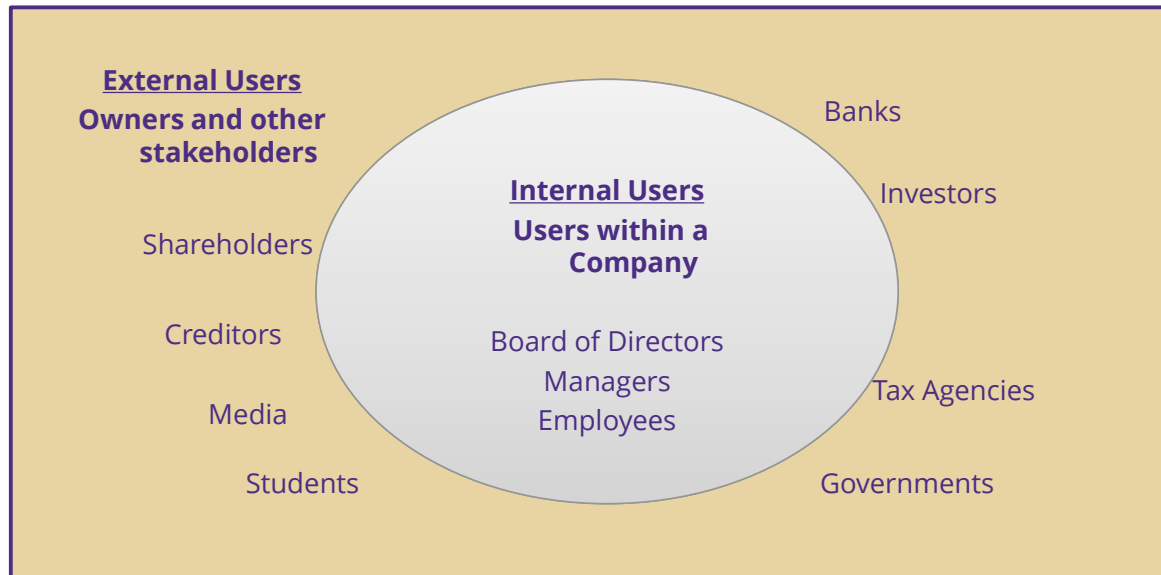
That same public company after IPO

CFO



Approx 450 employees

Accounting & Finance Function



Accounting & Finance Function

Data source for:

- Financial Reporting
- Accounting Systems
- FP&A
- Inventory management
- Supply Chain Mngt.
- And many more



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Transaction analysis

Transactions are events that affect the operations or finances of an organization.

- Accounting systems record transactions using a double-entry system using debits and credits.
- Transactions are recorded using the accounting equation:
 - **Assets = Liability + Equity**
- For example, increases to assets are debit entries

Transaction analysis

Transactions record the economic activities of the firm so that firms can report meaningful economic aggregates like net income, revenue, cost of goods/services sold, R&D expense, cash, inventory, what they owe (payables, debt), what they are owed (receivables, prepaid services).

Peach State University

We undertake an analysis of Peach State University Hotel (a real hotel's transactions with the name changed).

Three-part case with reference readings for each part.

Data analytics tool: Tableau

- Set-up in FSB remote labs using <https://fprod-hcs.foster.uw.edu/>
- In the u-drive use cmd to
- git clone <https://github.com/ashercurtis/datasets.io>

Peach State University

Part 1. Familiarity with the tool. Reference:

[Analytics_mindset_case_studies_PSU_Hotel_understanding_audit_analytics](#)

Part 2: Understanding sources of revenue. Reference:

[Analytics_mindset_case_studies_PSU_Hotel_Revenue](#)

Part 3: You will act as an advisor to identify events / transactions / transaction categories to craft the firm's non-GAAP earnings number. Reference:

[Analytics_mindset_case_studies_PSU_Hotel_background](#) (hint: search for events)

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Analyzing Public Disclosures

Accounting Income Matters!

Empirical Evidence suggests that accounting moves markets

Source: Ball and Brown, *Journal of Accounting Research*, 1968

EMPIRICAL EVALUATION OF ACCOUNTING INCOME NUMBERS

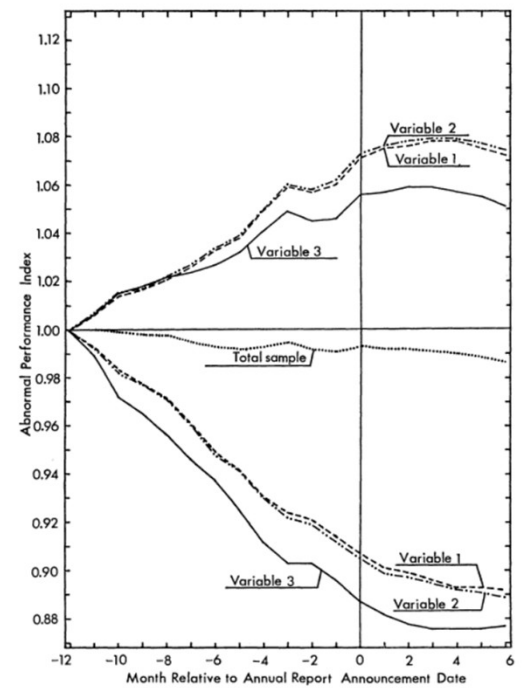


FIG. 1 Abnormal Performance Indexes for Various Portfolios

What is accounting income?

$$\text{Net Income} = \text{Revenues} - \text{Expenses}$$

For an accounting period (often reported quarterly, and annually)

The Income Statement

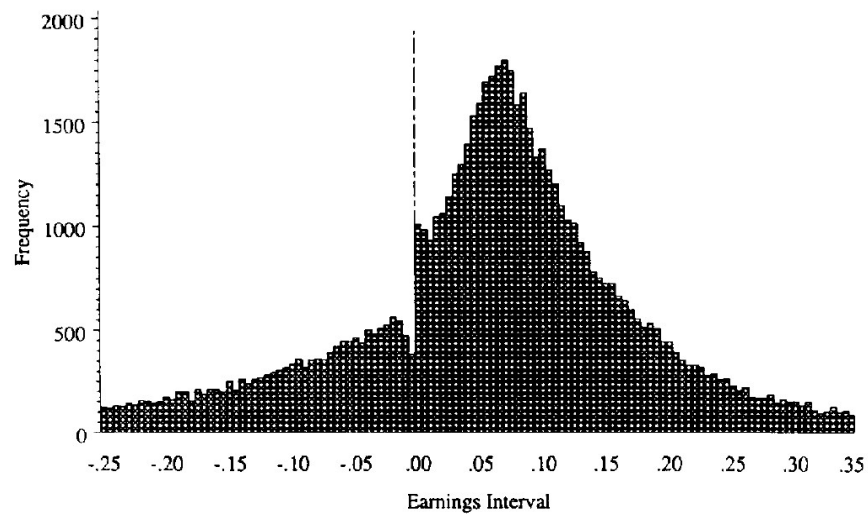
Displays financial performance derived from business operations over a specific accounting period by reporting:

- **Revenue:** inflows of economic benefits that increase shareholders equity.
 - e.g., sales revenue, service revenue, fees earned
- **Expenses:** Use or loss of economic benefits that decrease shareholders equity.
 - Incurred when you use resources to generate revenue, e.g., cost of goods sold, wages and salaries

If revenues are greater than expenses, a **net profit** is reported

If revenues are less than expenses, a **net loss** is reported

Analyzing Public Disclosures



Source: Burgstahler and Dichev, 1997 *J. of Accounting & Economics*

Analyzing Public Disclosures

Team Activity:

Part 1: Analysis of Marriott's Public Disclosures (as a class).

[Investor Relations | Marriott International](#)

Part 2: Analysis of other company filings (in teams)

Marriott

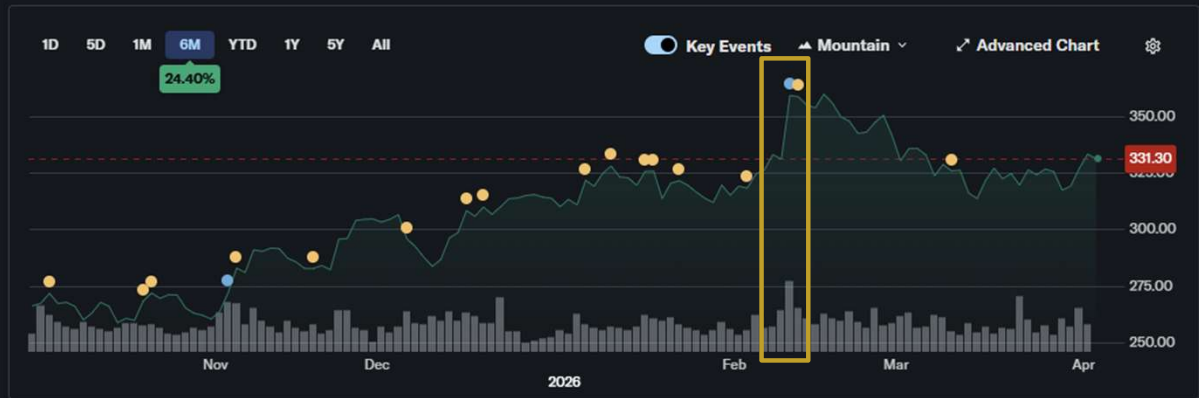
Marriott International, Inc. (MAR) ☆

331.30 -2.16 (-0.65%)

As of 3:27:36 PM EDT, Market Open.

Trade MAR on Coinbase

Trading disclosure ⓘ



GAAP and Non-GAAP

GAAP EPS

- ✓ Measured according to well established rules (GAAP)
- ✓ Often comparable across firms
- ✓ Includes transitory items, discontinued operations and extraordinary items

Pro-forma / Non-GAAP EPS

- ✓ Based on managers discretion
- ✓ Not often comparable across firms
- ✓ Highlighted by the SEC as a potential fraud risk
- ✓ Measures core earnings, / the firm's own measure of performance
- ✓ Often used in CEO compensation

Understanding the Income statement

Sales, COGS, SG&A, Depreciation

Interest, minority interests

Non-operating income, income tax

Special items

- restructuring charges

- write-offs

- M&A costs

Discontinued operations, extraordinary items

Most likely to recur

Least likely to recur

Frequency of Exclusions

Acquisitions	38%	Amortization of acquired intangibles	10%
Discontinued operations	15%	Capital charge	3%
Effects of accounting changes	7%	Effects of regulation changes	2%
Extinguishment	8%	Mark-to-market	3%
Foreign exchange gains and losses	9%	Tax adjustments	15%
Gains and losses	22%	Unclassified	46%
Impairment	16%		
In-process R&D	1%		
Other special items	6%		
Research and development	1%		
Restructuring	39%		
Settlements	15%		
Stock compensation	16%		
Write-downs	17%		

Source: "The Use of Adjusted Earnings in Performance Contracts"

Curtis, A., Li, V., and Patrick, P., (2021). *Review of Accounting Studies*, Vol. 26, pp. 1290–1322.

Submission

For today's class the required submission is in two parts submitted together in a word document.

1. Response to analyzing public disclosures
2. Response to analyzing transactions

Submission Part 1

Purpose: Extract examples from large public firms on the industry practice surrounding the reporting of earnings.

Add a header, “Analysis of Publicly Traded Hotel Industry Firms’ Earnings Disclosures” to your word document. Provide a summary and a screenshot of supporting evidence about earnings reporting.

Focus on whether they used GAAP or non-GAAP, and outline the exclusions, typically found for these hotels, and other unusual items you may find.

Analyzing Public Disclosures

Team Activity:

Part 2: Analysis of other company filings (in teams)

- [Hilton – Investor Relations](#)
- [Hyatt Hotels Corporation - Investor Relations](#)
- [Investors - InterContinental Hotels Group PLC](#)
- [Wyndham Hotels & Resorts, Inc. \(WH\)](#)

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Disclosure Decision

Data Analytics – Transaction Analytics

Purpose: Provide advisory services to Peach State University Hotels on how they should disclose their annual performance for the year.

Add a header, “Disclosure Advice for Peach State University Hotel” to your word document. Provide a summary and a screenshot of supporting evidence.

Refer to Part 1 as evidence from industry practice.

What will we do next?

The Accounting & Finance Function with Self-Service Analytics using Alteryx

- Explore some Compliance and Accounting & Finance Data Analytics from a production side.
- Use a self-service analytics tool (Alteryx) to produce working repeatable analytic end-to-end workflows for specific tasks.



Thank you

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